2N3055 (Hometaxial), 2N6253, 2N6254, 2N6371, 40251

Hometaxial-Base High-Power Silicon N-P-N Transistors

Rugged, Broadly Applicable Devices
For Industrial and Commercial Use

The RCA-2N3055 (Hometaxial)*, 2N6253, 2N6254 and 2N6371 are silicon n-p-n transistors intended for a wide variety of high-power applications. The hometaxial-base construction of these devices renders them highly resistant to second breakdown over a wide range of operating conditions.

These devices differ in maximum ratings for voltage and power dissipation. All are supplied in JEDEC TO-204MA hermetic steel packages.

The 2N3055 is also available in an epitaxial-base version. To obtain the hometaxial-base type described in this data sheet, order the 2N3055 (Hometaxial).

*Formerly 2N3055H.

Features:

- 2N6254: premium type from 2N3055 (Hometaxial) family
- Maximum safe-area-of-operation curves
- Low saturation voltages
- High dissipation capability
- Thermal-cycling rating curves

Applications:

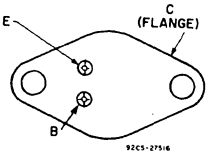
- Series and shunt regulators
- High-fidelity amplifiers
- Power-switching circuits
- Solenoid drivers
- Low-frequency inverters

MAXIMUM RATINGS, *Absolute-Maximum Values:*

	2N3055 (Hometaxial)	2N6253	2N6254	2N6371	40251	
* V_{CBO}	100	55	100	50	50	V
* $V_{CER(sus)}$ R_{BE} = 100 Ω	70	55	85	45	—	V
* $V_{CEO(sus)}$	60	45	80	40	40	V
$V_{CEV(sus)}$ V_{BE} = −1.5 V	90	55	90	50	50	V
* V_{EBO}	7	5	7	5	5	V
* I_C	15	15	15	15	15	A
* I_B	7	7	7	7	7	A
* P_T:						
≤25°C	115	115	150	117	117	W
>25°C	Derate linearly to 200°C					
* T_J, T_{stg}	−65 to +200					°C
* T_L: During soldering, at distances 1/32 in. (0.8 mm) from seating plane for 10 s max.	235					°C

*In accordance with JEDEC registration data formats JS-9 RDF-10; 2N3055 (Hometaxial) JS-6 RDF-2; 2N6253, 2N6254, 2N6371.

TERMINAL DESIGNATIONS



JEDEC TO-204MA
(See dimensional outline "A".)

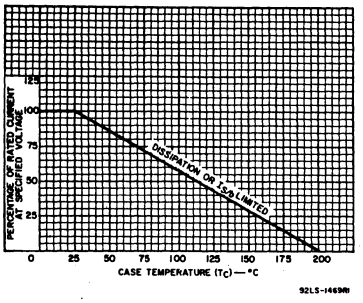


Fig. 1 — Current derating curve for all types.

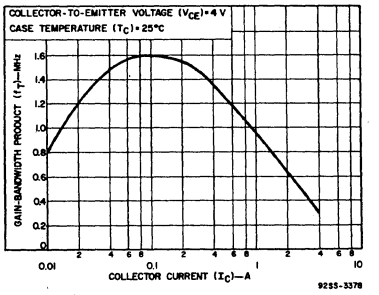


Fig. 2 — Typical gain-bandwidth product for all types.

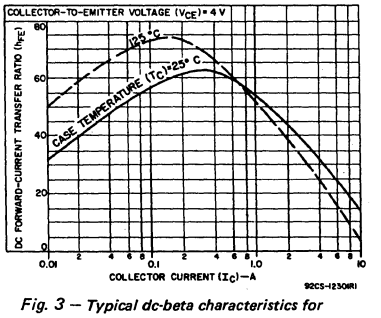


Fig. 3 — Typical dc-beta characteristics for 2N3055 (Hometaxial) and 2N6371.

2N3055 (Hometaxial), 2N6253, 2N6254, 2N6371, 40251

ELECTRICAL CHARACTERISTICS, *T_C = 25°C Unless Otherwise Specified.*

CHARACTERISTIC	TEST CONDITIONS VOLTAGE V dc		TEST CONDITIONS CURRENT A dc		LIMITS 2N3055 (Hometaxial)		2N6253		2N6254		2N6371		40251		UNITS
	V_{CE}	V_{BE}	I_C	I_B	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	
* I_{CEO}	25			0	–	–	–	1.5	–	–	–	1.5	–	–	mA
	30			0	–	0.7	–	–	–	–	–	–	–	–	
	60			0	–	–	–	–	–	1	–	–	–	–	
* I_{CEX}	40	−1.5			–	–	–	–	–	–	–	–	–	2	mA
	45	−1.5			–	–	–	–	–	–	–	2	–	–	
	55	−1.5			–	–	–	2	–	–	–	–	–	–	
	100	−1.5			–	5	–	–	–	0.5	–	–	–	–	
* T_C = 150°C	40	−1.5			–	–	–	–	–	–	–	10	–	10	
	50	−1.5			–	–	–	10	–	–	–	–	–	–	
	100	−1.5			–	30	–	–	–	5	–	–	–	–	
* I_{EBO}		−5			–	–	–	10	–	–	–	10	–	10	mA
		−7			–	5	–	–	–	0.5	–	–	–	–	
$V_{(BR)CBO}$			0.1		–	–	–	–	–	–	–	–	50	–	V
$V_{(BR)CEV}$		−1.5	0.1		–	–	–	–	–	–	–	–	50	–	V
$V_{(BR)EBO}$ I_E = 0.01 A			0		–	–	–	–	–	–	–	–	5	–	V
* $V_{CEO(sus)}$			0.2[a]	0	60	–	45	–	80	–	40	–	40	–	V
* $V_{CER(sus)}$ R_{BE} = 100 Ω			0.2[a]		70	–	55		85	–	45	–	–	–	
$V_{CEV(sus)}$		−1.5	0.1[a]		90	–	55	–	90	–	50	–	–	–	
* h_{FE}	4		3[a]		–	–	20	70	–	–	–	–	–	–	
	4		4[a]		20	70	–	–	–	–	–	–	–	–	
	2		5[a]		–	–	–	–	20	70	–	–	–	–	
	4		8[a]		–	–	–	–	–	–	15	60	15	60	
	4		10[a]		5	–	–	–	–	–	–	–	–	–	
	4		15[a]		–	–	3	–	5	–	–	–	–	–	
	4		16[a]		–	–	–	–	–	–	4	–	–	–	
* V_{BE}	4		3[a]		–	–	–	1.7	–	–	–	–	–	–	V
	4		4[a]		–	1.8	–	–	–	–	–	–	–	–	
	2		5[a]		–	–	–	–	–	1.5	–	–	–	–	
	4		8[a]		–	–	–	–	–	–	–	–	–	2.2	
	4		16[a]		–	–	–	–	–	–	–	4	–	–	
* $V_{CE(sat)}$			3[a]	0.3[a]	–	–	–	1	–	–	–	–	–	–	V
			4[a]	0.4[a]	–	1.1	–	–	–	–	–	–	–	–	
			5[a]	0.5[a]	–	–	–	–	–	0.5	–	–	–	–	
			8[a]	0.8[a]	–	–	–	–	–	–	–	1.5	–	1.5	
			10[a]	3.3[a]	–	8	–	–	–	–	–	–	–	–	
			15[a]	3[a]	–	–	–	–	–	4	–	–	–	–	
			15[a]	5[a]	–	–	–	4	–	–	–	–	–	–	
			16[a]	4[a]	–	–	–	–	–	–	–	4	–	–	

* In accordance with JEDEC registration data formats JS-9 RDF-10: 2N3055H; JS-6 RDF-2: 2N6253, 2N6254, 2N6371.

[a] Pulsed: Pulse duration = 300 μs, duty factor = 1.8%.

2N3055 (Hometaxial), 2N6253, 2N6254, 2N6371, 40251

ELECTRICAL CHARACTERISTICS, $T_C = 25°C$ *Unless Otherwise Specified.*

CHARACTERISTIC	TEST CONDITIONS				LIMITS										UNITS
	VOLTAGE V dc		CURRENT A dc		2N3055 (Hometaxial)		2N6253		2N6254		2N6371		40251		
	V_{CE}	V_{BE}	I_C	I_B	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	
* h_{fe} f = 1 kHz	4		1		15	120	10	–	10	–	10	–	–	–	
f_T	4		1		800	–	–	–	–	–	800	–	–	–	kHz
* $\|h_{fe}\|$ f = 0.4 MHz	4		1		–	–	2	–	2	–	2	–	–	–	
* f_{hfe}	4		1		10	–	10	–	10	–	–	–	–	–	kHz
$I_{S/b}$	39				–	–	–	–	–	–	–	–	3	–	
t_p = 1 s	40				2.9	–	–	–	–	–	2.9	–	–	–	
nonrep.	45				–	–	2.55	–	–	–	–	–	–	–	A
	60				1.95	–	–	–	–	–	–	–	–	–	
	80				–	–	–	–	1.87	–	–	–	–	–	
$R_{\theta JC}$					–	1.5	–	1.5	–	1.17	–	1.5	–	1.5	°C/W

*In accordance with JEDEC registration data formats JS-9 RDF-10: 2N3055H; JS-6 RDF-2: 2N6253, 2N6254, 2N6371.

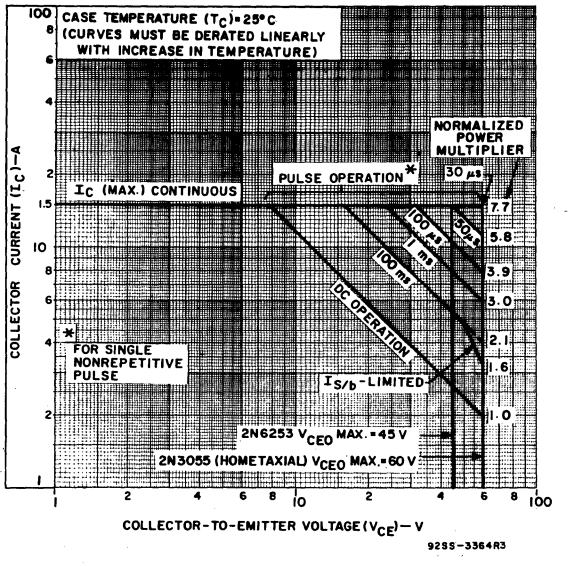


Fig. 4 – Maximum operating areas for 2N6253 and 2N3055 (Hometaxial).

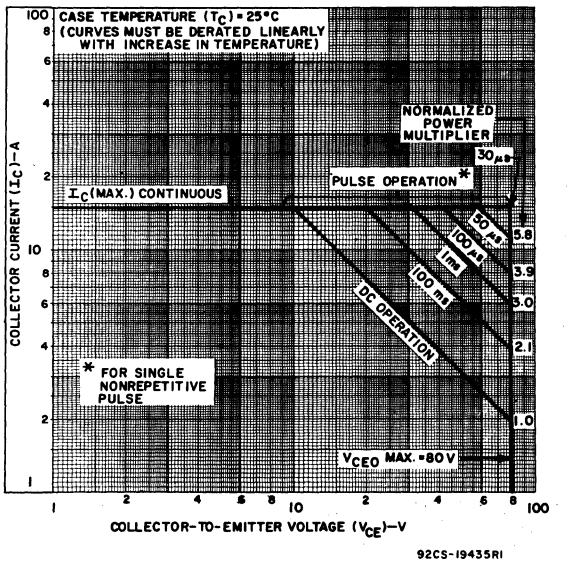


Fig. 5 – Maximum operating areas for 2N6254.

2N3055 (Hometaxial), 2N6253, 2N6254, 2N6371, 40251

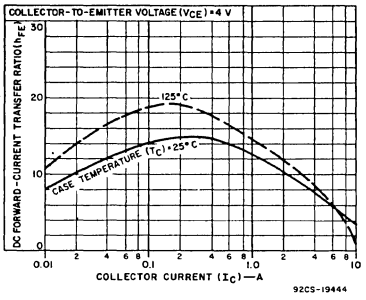


Fig. 6 – Typical dc-beta characteristics for 2N6253.

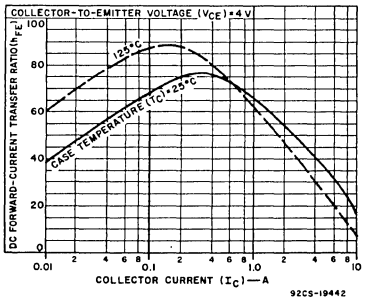


Fig. 7 – Typical dc-beta characteristics for 2N6254.

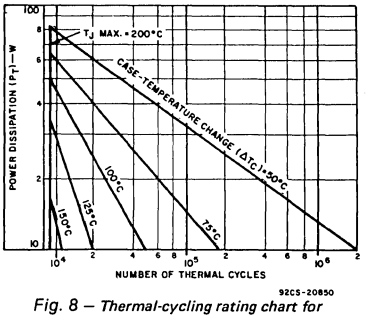


Fig. 8 – Thermal-cycling rating chart for 2N3055 (Hometaxial), 2N6253 and 2N6371.

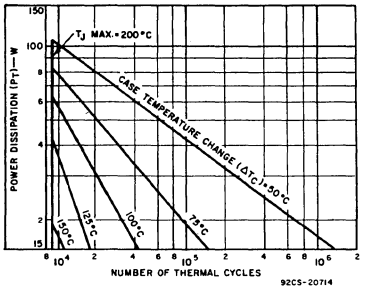


Fig. 9 – Thermal-cycling rating chart for 2N6254.

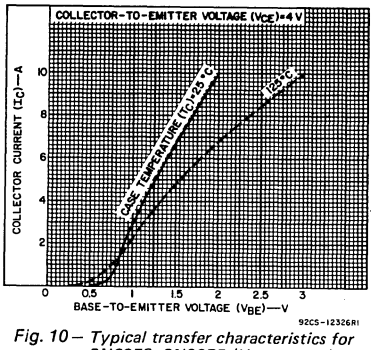


Fig. 10 – Typical transfer characteristics for 2N6253, 2N3055 (Hometaxial), 2N6371 and 40251.

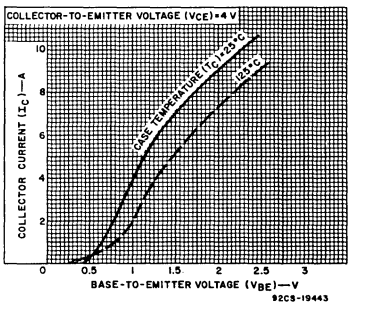


Fig. 11 – Typical transfer characteristics for 2N6254.

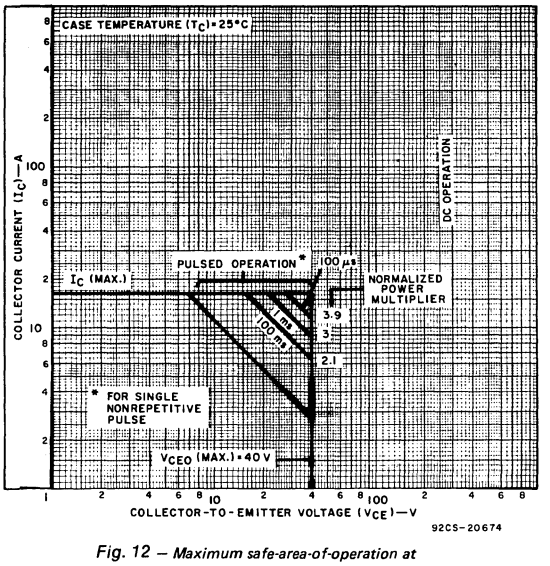


Fig. 12 – Maximum safe-area-of-operation at case temperature of 25° C for 2N6371.

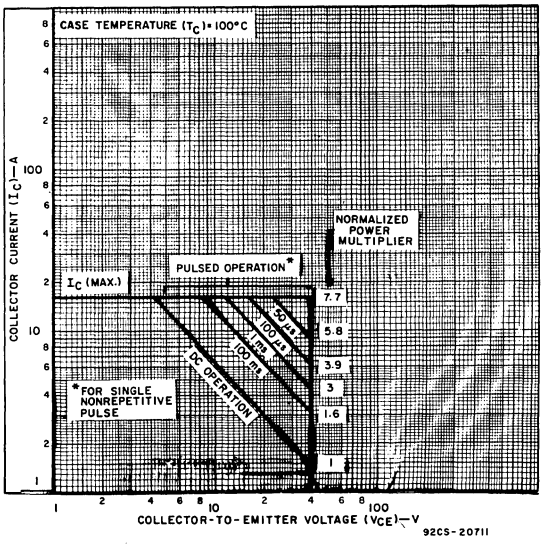


Fig. 13 – Maximum safe-area-of-operation at case temperature of 100° C for 2N6371.

2N3055 (Hometaxial), 2N6253, 2N6254, 2N6371, 40251

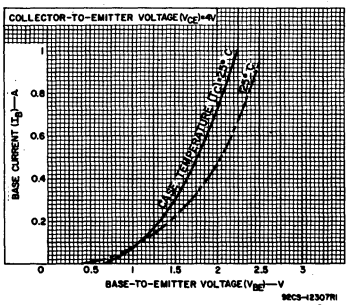


Fig. 14 — Typical input characteristics for 2N3055 (Hometaxial), 2N6371 and 40251.

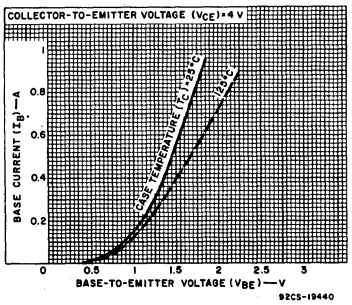


Fig. 15 — Typical input characteristics for 2N6253.

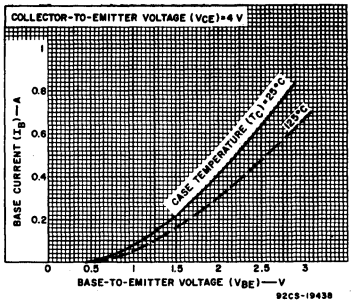


Fig. 16 — Typical input characteristics for 2N6254.

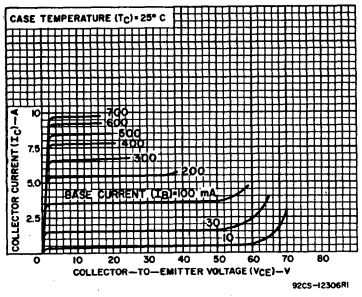


Fig. 17 — Typical output characteristics for 2N3055 (Hometaxial) and 2N6371.

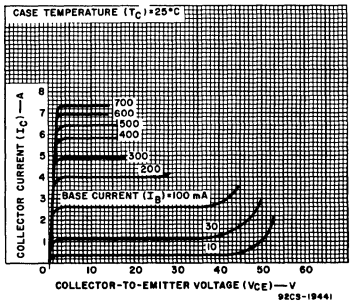


Fig. 18 — Typical output characteristics for 2N6253.

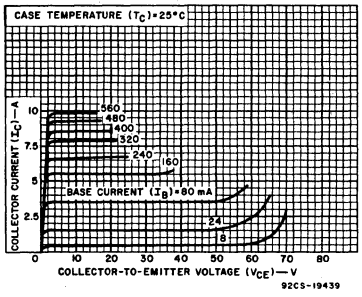


Fig. 19 — Typical output characteristics for 2N6254.

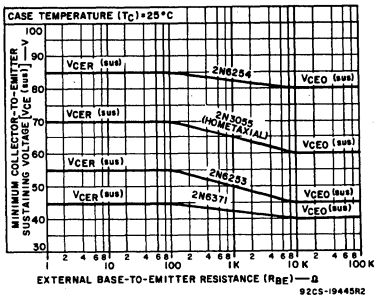


Fig. 20 — Sustaining voltage vs. base-to-emitter resistance for all types.